LearnLens AI

Personalized Tutoring & Adaptive Learning — Comprehensive Operational Guide

Hackathon Track: Track D (Personalized Tutoring & Adaptive Learning) | **Multimodal AI Hackathon 2026**

GitHub Repository: https://github.com/rahulgoreofficial/LearnLens-AI-Personalized-Tutoring-Adaptive-Learning

Executive Summary: LearnLens AI is a functional, full-stack educational AI companion that unifies disparate study assets (PDF textbooks, PowerPoint slides, video subtitles, and YouTube lectures) into a source-cited multimodal knowledge base. It delivers: **(1) Verifiable Provenance Citations** identifying exact PDF pages, PowerPoint slides, or video timestamps; **(2) Grounded AI Tutoring** with anti-hallucination guardrails; **(3) Dynamic Assessment Generation** across 3 difficulty tiers; and **(4) Transparent Adaptive Mastery Tracking**.

1. Problem Statement & Platform Vision

Students are overwhelmed by fragmented learning media: textbook PDFs, lecture slide decks (.pptx), recorded lectures (.srt subtitles), and online YouTube videos. Generic conversational LLMs frequently invent facts, generate generic quizzes detached from assigned course materials, and lack actionable guidance on which exact slide or book page to revisit.

The LearnLens Solution: An end-to-end retrieval-augmented system that ingests multimodal documents, preserves physical provenance tags on every text chunk, feeds verified excerpts to Gemini (via the official google-genai SDK), and generates verifiable citations so students can expand and inspect the exact source text.

2. System Architecture & Tech Stack

Component Layer	Technologies & Architecture Description
Frontend UI	React 19, TypeScript, Vite, Tailwind CSS v4, Lucide Icons. Features responsive Dark Theme, Interactive Citation Cards, Chunk Inspector, and Mastery Gauges.
Backend API	Python 3.14 / 3.11, FastAPI, Uvicorn, Pydantic v2. Clean REST architecture with CORS & validation.
Multimodal Parsers	**PyMuPDF** (extracts text with 1-based page numbers), **python-pptx** (extracts slides, tables & speaker notes), **webvtt-py / SRT regex** (timestamps), and **youtube-transcript-api**.
Retrieval Engine	TF-IDF Vectorizer with n-grams (1,2), sublinear term frequency scaling, keyword boosting, and cosine similarity ranking. Purely deterministic and zero external vector DB dependencies.
AI Model & SDK	Google Gemini 2.5 Flash via official google-genai Python SDK. Strictly grounded prompting.

3. Detailed Feature Breakdown

• **Feature 1: Multimodal Study Material Library:** Allows students to drag-and-drop PDF textbooks, PowerPoint slide decks, subtitle transcripts (.srt/.vtt), and markdown notes. Also includes a direct YouTube transcript importer that takes any video URL and captures timestamped transcript blocks. The built-in **Chunk Inspector Modal** lets students inspect exactly how each page or slide was parsed into searchable chunks.

• **Feature 2: Source-Grounded AI Tutor:** When a student asks a question, the retrieval engine searches all indexed chunks. If similarity meets relevance thresholds, the top chunks are formatted into a structured prompt for Gemini. The tutor returns answers citing sources in-text (e.g. [Transformer Slides, Slide 3]) paired with interactive **Citation Cards**. Clicking any card expands the verified excerpt. If no relevant chunks exist, strict guardrails instruct the tutor to state that the library lacks sufficient context.

• **Feature 3: Personalized Dynamic Quizzes:** Synthesizes grounded Multiple Choice Questions (MCQs) dynamically from the student's study materials across Easy, Medium, and Hard tiers. The interactive quiz player features real-time answer checking, full answer explanations, and citations to the supporting document page or slide.

• **Feature 4: Adaptive Mastery Engine:** Empirically monitors student recall per topic. Calculates topic mastery using a transparent 70/20/10 formula: **70% Base Accuracy**, **20% Recency Momentum** (past 3 attempts), and **10% Difficulty Factor** (bonuses for hard questions). Identifies weak topics (< 60% mastery) and provides one-click targeted practice.

• **Feature 5: Central Study Dashboard:** Real-time mission control displaying materials count, chunks indexed, questions asked, quiz accuracy %, average mastery %, weak topic alerts, and an adaptive study queue recommending specific pages and slides to review.

4. The Transparent Topic Mastery Scoring Algorithm

Mastery(Topic) = min(100, max(0, Base_Accuracy * 0.70 + Recency_Momentum * 0.20 + Difficulty_Bonus * 0.10))

• **70% Base Accuracy:** Evaluates raw performance as *(Correct Answers / Total Attempted) * 70*.
• **20% Recency Momentum:** Measures active learning progress by weighting the student's 3 most recent attempts.
• **10% Difficulty Factor:** Rewards successful retention on analytical Medium (0.5x) and Hard (1.0x) questions.
• **Classification Bands: Mastered** (≥ 80%), **In Progress** (50% – 79%), and **Needs Practice** (< 50%).

5. Step-by-Step Guide: How to Start the Program

Follow these exact commands to install, configure, and launch the complete working application:

Step 5.1: Clone the Repository

```
git clone https://github.com/rahulgoreofficial/LearnLens-AI-Personalized-Tutoring-Adaptive-Learning.git
cd LearnLens-AI-Personalized-Tutoring-Adaptive-Learning
```

Step 5.2: Install Backend & Frontend Dependencies

```
# 1. Install Backend Dependencies (FastAPI, PyMuPDF, python-pptx, google-genai, scikit-learn)
pip install -r requirements.txt

# 2. Install Frontend Dependencies (React, TypeScript, Vite, Tailwind CSS v4, Lucide)
cd frontend
npm install
cd ..
```

Step 5.3: Environment Variable Configuration

Create a .env file in the project root or backend directory by copying .env.example:

```
copy .env.example .env

# Content of .env:
GEMINI_API_KEY=your_gemini_api_key_here
GEMINI_MODEL=gemini-2.5-flash
```

Step 5.4: Launch the Backend & Frontend Servers

Run the backend API and frontend dev server in two separate terminals:

```
--- TERMINAL 1 (FastAPI Backend Server) ---
python backend/run.py
# Output: Uvicorn running on http://127.0.0.1:8000 (Swagger docs at http://127.0.0.1:8000/docs)

--- TERMINAL 2 (Vite + React Frontend Server) ---
cd frontend
npm run dev
# Output: VITE ready in 329 ms. Local: http://localhost:5173/
```

Step 5.5: Automated Verification Suite

Execute backend/test_api.py to test all 6 core endpoints end-to-end:

```
python backend/test_api.py

# Verifies:
# [1] Health check & model status  [2] Multimodal material & chunk ingestion
# [3] Grounded retrieval & citations [4] Dynamic quiz question synthesis
# [5] Quiz grading & telemetry     [6] Adaptive mastery progress calculation
```

6. Interactive Demonstration Walkthrough (For Hackathon Judges)

Follow this 5-stage demonstration script to present all features of LearnLens AI in under 3 minutes:

• **Stage 1: Instant Multimodal Seeding:** Open http://localhost:5173/. Click the **'Load Demo Materials'** button in the top navigation. In under 1 second, 4 multimodal course assets are indexed: a multi-page PDF textbook, PowerPoint slide deck, an SRT lecture subtitle file, and Markdown notes.

• **Stage 2: Inspecting Document Provenance:** Navigate to **'Study Library'**. Observe the material cards displaying total pages, slides, and timestamps. Click **'Inspect Chunks'** on *'Transformer Architecture Slides'* to demonstrate that each slide was segmented into a discrete chunk with its exact locator ('Slide 1' to 'Slide 4').

• **Stage 3: Grounded Tutoring & Verifiable Citations:** Navigate to **'AI Tutor'**. Click the quick prompt: *'What is the formula for scaled dot-product attention and why is the scaling factor used?'*. When the tutor answers, point out the citation badge (Transformer Architecture Slides - Slide 3). Click the card to expand and reveal the verified source excerpt.

• **Stage 4: Adaptive Quiz & Grounded Explanations:** Navigate to **'Quizzes'**. Click **'Generate Adaptive Quiz'**. Select an option and click **'Check Answer'**. Point out the instant green/red indicator, the pedagogical explanation, and the verified source citation link. Finish the quiz to see the mastery gains.

• **Stage 5: Adaptive Mastery Telemetry:** Navigate to **'Adaptive Mastery'**. Review the color-coded mastery cards. Click **'How Mastery is Calculated'** to review the transparent 70/20/10 formula. Click **'Practice'** next to a weak topic to immediately launch a targeted quiz.

7. AI-Assisted Development Disclosure & Compliance

In compliance with the Multimodal AI Hackathon 2026 rules, we disclose that **Google Antigravity** and the **Gemini Coding Model** were utilized during AI-assisted development for:

- **Architecture & Schemas:** Designing Pydantic models for DocumentChunk, Citation, and Quiz.
- **Multimodal Parsing:** Format dispatch logic for PyMuPDF, python-pptx, and transcript parsers.
- **Grounded Prompt Engineering:** System instructions enforcing strict verifiable provenance citations.
- **Heuristic Telemetry:** Mathematical formulation of the transparent 70/20/10 mastery algorithm.

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